



TEST CODE **02220020**

MAY/JUNE 2006

FORM TP 2006209

CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE

UNIT 2: AGRICULTURE, ENERGY AND ENVIRONMENTAL POLLUTION

PAPER 02

2 hours 30 minutes

26 MAY 2006 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of **NINE** questions.
2. Answer **SIX** questions.
3. Write your answers in the answer booklet provided.
4. The use of a non-programmable calculator is allowed.

MODULE 1

Answer ANY TWO questions.

1. (a) Identify TWO forms of soil degradation associated with excessive irrigation. [2 marks]
- (b) Explain how EACH form of soil degradation identified in (a) occurs. [16 marks]
- (c) Recommend ONE sustainable practice in agriculture to address EACH form of degradation identified in (a) above. [2 marks]
- Total 20 marks**
2. (a) What is meant by 'mechanization of agriculture'? [2 marks]
- (b) Describe FIVE factors that limit the mechanization of agriculture in the Caribbean. [10 marks]
- (c) Justify the following statement.
- "Agroecosystems are different from natural unmanaged ecosystems."
- [8 marks]
- Total 20 marks**
3. (a) Distinguish between 'mariculture' and 'aquaculture'. [2 marks]
- (b) Describe THREE management techniques that are used for increasing the yield of aquaculture. [12 marks]
- (c) Discuss TWO environmental issues resulting from the practice of aquaculture in the Caribbean. [6 marks]
- Total 20 marks**

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MODULE 2

Answer ANY TWO questions.

4. Figures 1 and 2 are provided as an insert.

Figure 1 shows the projected annual demands for electricity in the Caribbean, and Figure 2 shows the projected annual world production of oil.

- (a) Considering predictions based on Figures 1 and 2, suggest the impact on Caribbean countries if the projected change in annual demand for electricity and oil production is realized.
[6 marks]
- (b) Discuss TWO alternative energy sources that Caribbean countries can exploit to address the impact identified in (a) above. (Include in your response the advantages and disadvantages of exploiting both energy sources.)
[14 marks]

Total 20 marks

5. (a) Explain the function of EACH of the following in a nuclear energy plant:

- (i) Control rods [5 marks]
- (ii) Primary water circuit [4 marks]

(b) Evaluate the recommendation below.

“Caribbean countries should satisfy their growing demand for electricity through the use of nuclear energy rather than building new conventional fossil fuel power plants.”

(Include in your response the advantages and disadvantages of both methods of producing electricity.)
[11 marks]

Total 20 marks

6. (a) Using a suitable example in each case, distinguish between ‘energy conservation’ and ‘energy efficiency’.
[4 marks]

(b) Evaluate the statement below.

“The implementation of measures to conserve and use energy more efficiently is the best and cheapest means of making more energy available.”
[6 marks]

(c) Discuss TWO ways in which the transportation sector in your country could conserve energy.
[10 marks]

Total 20 marks

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MODULE 3

Answer ANY TWO questions.

7. Figure 3 shows the variation in coral cover between 1970 and 2000 for a Caribbean island, Z. Tests on coastal waters indicate high levels of turbidity and faecal coliform.

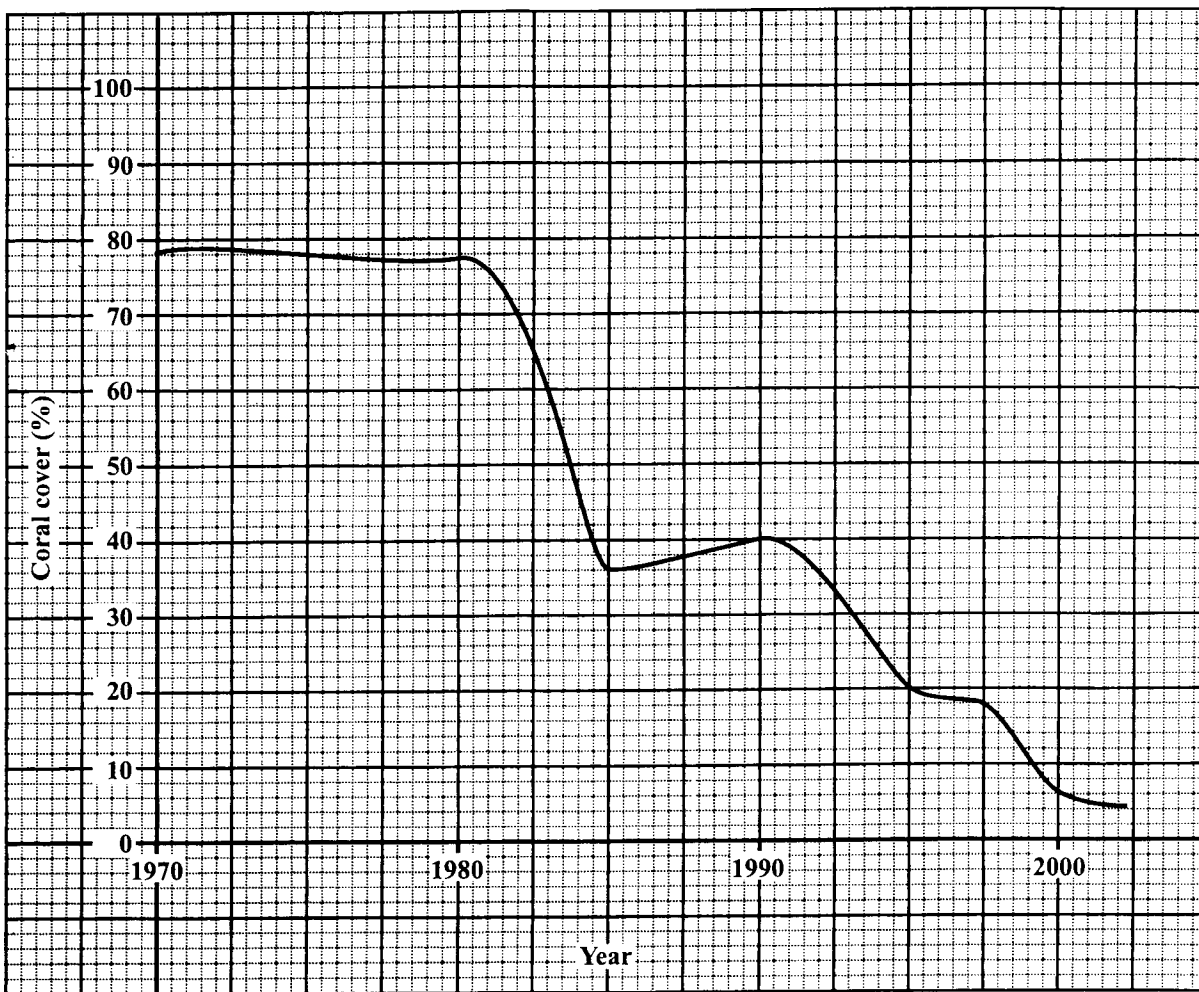


Figure 3. Variation in coral cover

- (a) Outline the trend shown in Figure 3. [3 marks]
- (b) Identify the TWO **major types** of water pollutant existing in the coastal waters of Island Z. [2 marks]
- (c) For EACH of the pollutants identified in (b) above, state
- (i) its nature [2 marks]
 - (ii) its source. [2 marks]
- (d) Explain how the loss of coral cover could be caused by ONE of the pollutants identified. [4 marks]
- (e) Outline a water treatment process that would address the high levels of faecal coliform in coastal waters. [7 marks]

Total 20 marks

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8. (a) What is a 'refuse derived fuel' incinerator? [2 marks]
- (b) Discuss the use of incinerators as a MAJOR method of solid waste disposal in Caribbean countries. (In your response, consider at least TWO advantages and THREE disadvantages.) [18 marks]

Total 20 marks

9. (a) Top Hill game reserve is open for bird hunting for three months each year beginning in May. Soil samples taken from different areas of the reserve in April were tested and found to have small amounts of the hazardous chemical pesticide DDT.
- (i) Would you recommend that humans eat the game birds that are caught? Justify your answer. [8 marks]
- (ii) Describe ONE technological process that can be used to clean up soil contaminated with hazardous waste. [4 marks]
- (b) (i) What is the 'Basel Convention'? (Include at least TWO of its features in your response.) [3 marks]
- (ii) Explain why the Basel Convention was developed. [5 marks] '

Total 20 marks

END OF TEST

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Insert for Question 4

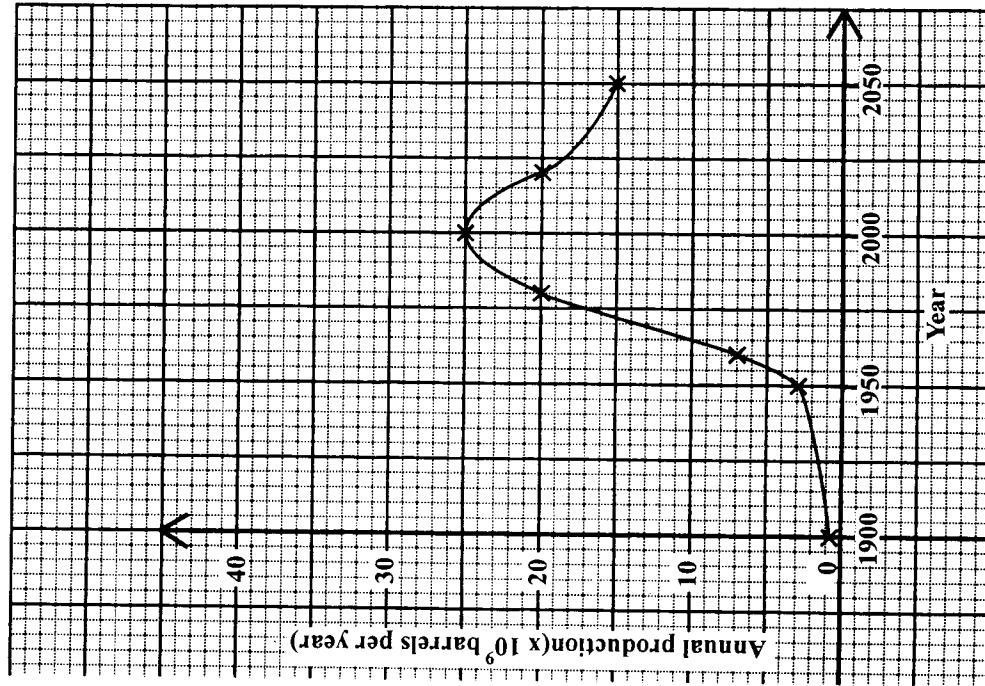


Figure 2. Projected annual world production of oil

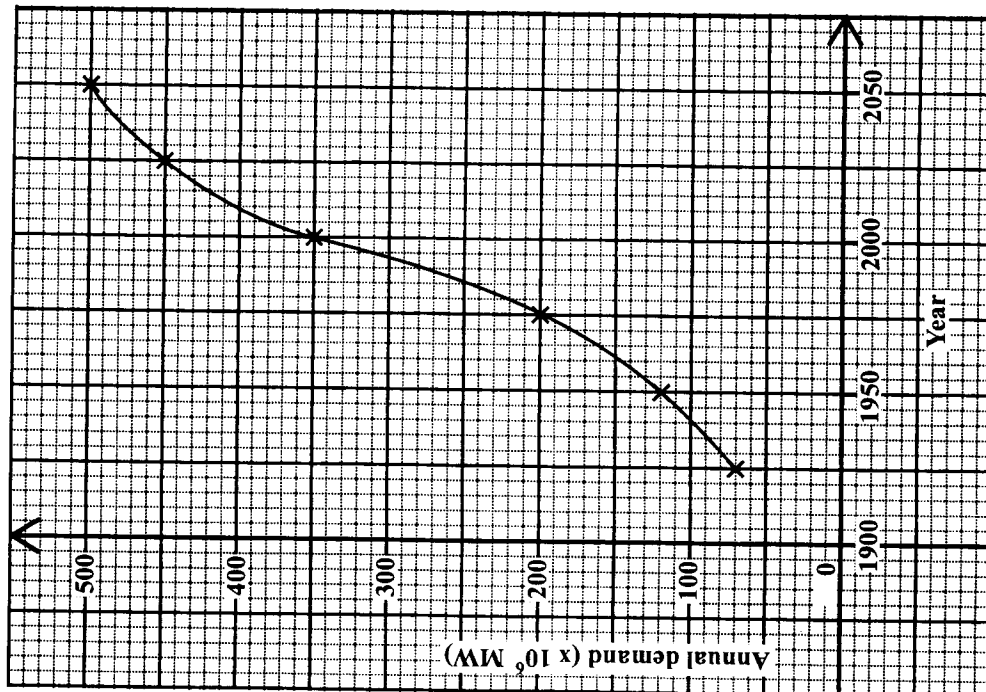


Figure 1. Projected annual demand for electricity in the Caribbean